

See risk before it happens

IT Consultant Challenge 2026 · Round 1 · Company A

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1. Identify – operations and pain points
2. Prioritise – why delivery risk first
3. Solve – solution and 12-week MVP
4. Scale – rollout to SE, NO, DK

Current operating context

COMPANY A

A leading Danish building-materials retailer in Denmark, Norway and Sweden

CUSTOMERS
Construction businesses and private customers

FOOTPRINT
Hundreds of stores

MARKETS
Operational needs differ between Denmark, Norway and Sweden

TECHNOLOGY LANDSCAPE

ERP

Role: Manages orders, inventory and inbound and outbound goods movements.

Explain delivery risk meaning
- Is stock confirmed?
- Is the order picked?

Gap
Is status updated in real time or in batches?

TMS

Role: Transport Management System for delivery planning and tracking.

Explain delivery risk meaning
- Planned slot, vehicle, carrier status

Gap
One shared TMS, or one per country?

POWER BI

Role: Dashboards for order status and operational performance.

Delivery risk meaning
- What has already happened

Gap
Backward-looking: no per-order risk, no ranking

GOOGLE MAPS

Role: Used to estimate travel time.

Delivery risk meaning
- Only the transport leg

Gap
Ignores stock, picking, dispatch and vehicle availability

Each system sees one piece of the order. None of them answers “Which orders will be late, and what should we do now?” That gap is where our solution acts.

Four pain points in operations

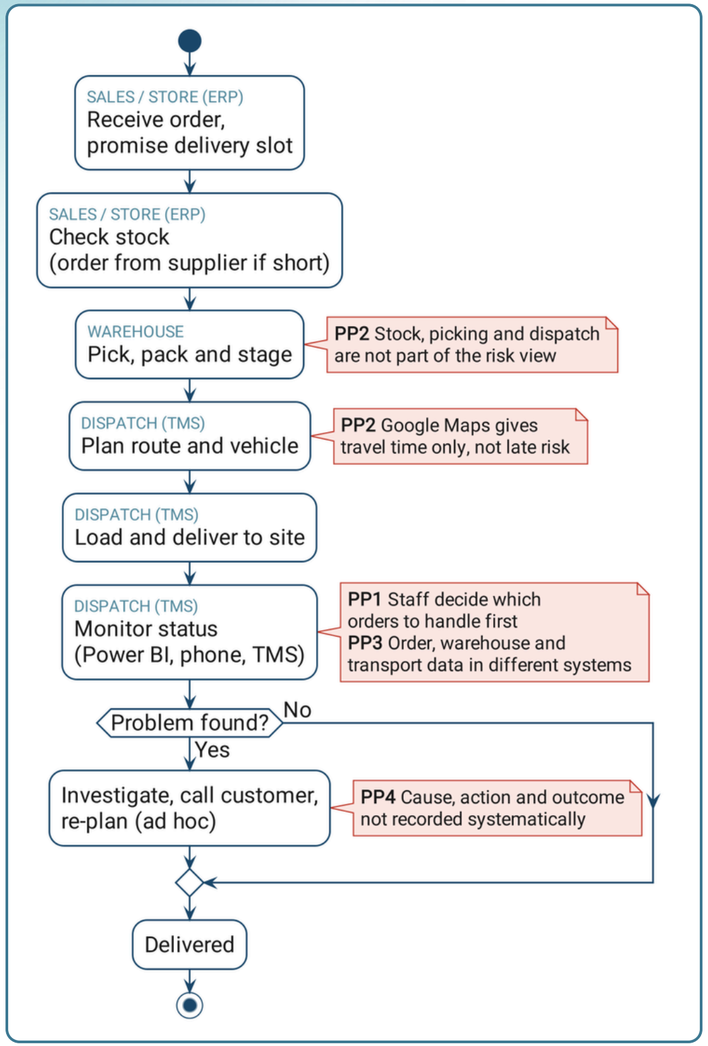


Figure 1: Today's order-to-delivery flow (as-is)

	PAIN POINT	IMPACT	TO VERIFY
1	**Hard to prioritise urgent orders** Dashboards show status and totals, so staff still decide by hand which order to handle first.	More manual work, slower response, missed cases	Do staff prioritise by hand today?
2	**Late-delivery risk not fully assessed** Google Maps gives travel time, but risk also depends on stock, picking, dispatch, vehicles and carriers.	Risk is found too late to change the plan	Is lateness predicted per order, or only ETA?
3	**No end-to-end view of risk** Order, warehouse and transport data sit in different systems and update at different times.	More checking, slower exception handling	Do systems share data often enough?
4	**Late-case data is not reused** Causes, actions and outcomes are not recorded, so nobody knows which fixes work.	Problems repeat, forecasts cannot improve	Are causes and outcomes recorded today?

Risk is visible only once a delivery is already in trouble.

Common root cause Company A manages delivery by looking backwards. Each system shows what has happened, not which orders will be late, so problems surface only after the customer is affected

The real problem, tackled first

Consolidated business problem

Company A may already have stable systems for orders, transport planning and reporting. What it lacks is one predictive capability that finds orders at risk of late delivery, ranks them by urgency and supports timely action. The opportunity is not another dashboard or routing tool, but proactive, risk-based order management.

01

Customers feel it directly

A late delivery can stop a building site. Reliability decides where contractors buy next.

02

Biggest impact in Sweden

The brief says delivery matters most in Sweden, so a pilot there shows value fastest.

03

Helps the other challenges

One data layer eases fragmented systems. A ranked worklist cuts manual checking.

04

Fast to market

A layer on top of ERP, TMS and Power BI. Nothing has to be replaced first.

The four challenges compared (impact / effort)

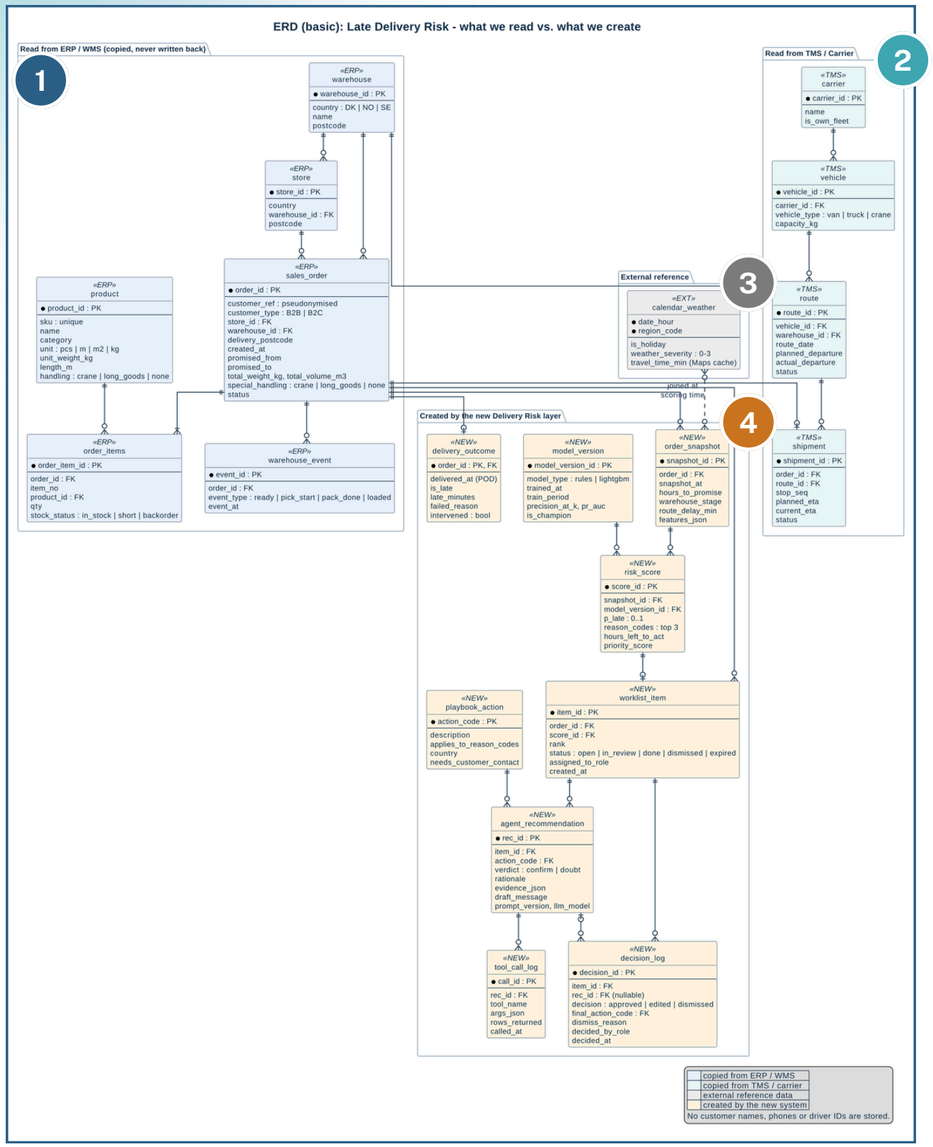
Late-delivery risk: high / medium ✔ first

Manual processes: medium / low → next

Store operations: medium, most urgent in NO / high → later

Fragmented systems: high, long-term / very high → not an MVP

Data model



1 **Read from ERP / WMS** 6 tables, copied every 15–60 min

promised_from / _to on sales_order: the slot we promised. It defines “late”, so it is the label every model learns from.
warehouse_event (ready, pick_start, pack_done, loaded): a slow pick shows up hours before the truck leaves.
order_items + **product**: stock status per line and handling (crane, long goods). **country** on every location; **customer_ref** pseudonymised.

2 **3** **Read from TMS / carrier + external** “Will the truck make it?”

carrier › vehicle › route › shipment › sales_order

Planned vs actual departure and current ETA feed reason codes such as VEHICLE_DELAYED. No driver column, by design.
calendar_weather: holiday, weather severity 0–3, cached Maps travel time per hour and region, joined at scoring time.

4 **Created by the Delivery Risk layer** 9 new tables, every step leaves a row

PREDICT ›

`order_snapshot`
`model_version`
`risk_score`

Scored on what was known at that hour. No look-ahead; any flag can be replayed.

ACT ›

`worklist_item`
`playbook_action`
`agent_recommendation`
`tool_call_log`

The agent may only propose an action listed in playbook_action.

DECIDE ›

`decision_log`

Works with the agent switched off. A role is stored, never a name.

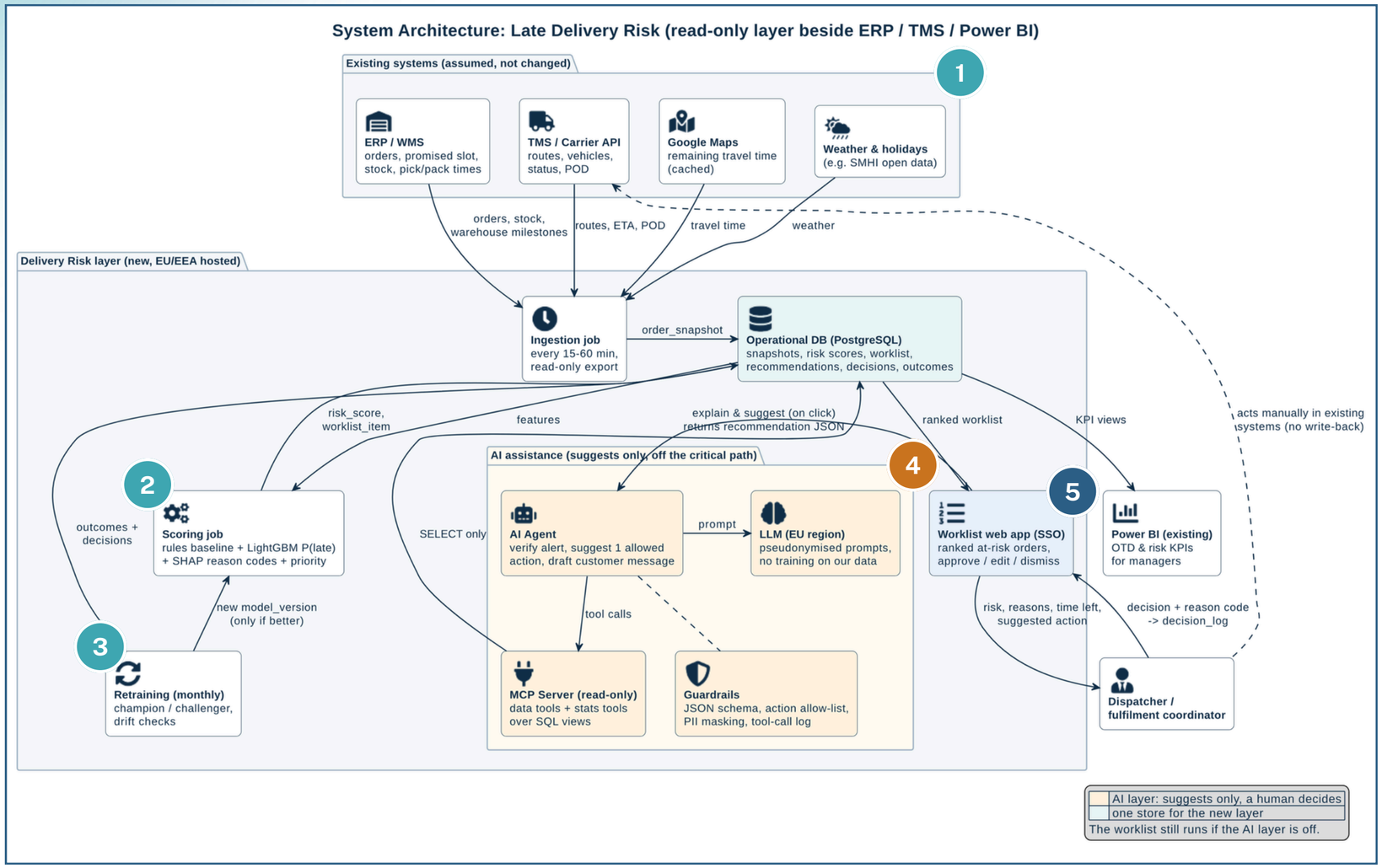
LEARN

`delivery_outcome`

intervened = true keeps rescued orders out of the “safe” examples.

5

System architecture



If the AI layer is off, the worklist still runs. ERP and TMS stay the system of record.

1 Four feeds in, nothing written back

ERP / WMS, TMS / Carrier API, Google Maps (cached) and weather & holidays. Every 15–60 min the ingestion job writes one **order_snapshot** per open order. File export (MVP) › read replica (pilot) › event stream (scale).

2 Critical path

Ingest › PostgreSQL › scoring › worklist. Rules first, LightGBM beside them; SHAP gives three reason codes. Priority = P(late) × impact × urgency.

3 Learning loop

Monthly retrain on outcomes and decisions. Champion vs challenger, drift checks.

4 AI assistance, off the critical path

The agent checks the alert, proposes one action from **playbook_action**, drafts the customer message in sv / no / da. It reads only through a **MCP server** (country-scoped, ≤ 50 rows, every call logged). LLM in an EU region on pseudonymised prompts; JSON schema and allow-list guardrails.

5 People at the edge

Dispatcher works a short, capped worklist behind SSO: approve, edit or dismiss, then acts in TMS or ERP as today. Managers keep Power BI with new KPI views.

6

MCP server

One read-only MCP server, fourteen small tools over SQL views. The agent can look, never touch.

Data tools what is true about this order right now

`get_order_context`	Promise, lines, stock and warehouse progress for one order
`get_risk_explanation`	Latest p_late, top-3 reason codes, rules fired, score trend
`get_shipment_status`	Route, vehicle, current ETA, delay of earlier stops
`list_high_risk_orders`	Ranked open worklist for one warehouse
`get_vehicle_options`	Other vehicles with free capacity that arrive in time
`get_alt_warehouse_stock`	Other DCs or stores that can supply short lines in time
`get_similar_past_cases`	Past cases with the same reasons: action taken, outcome
`get_playbook`	The allow-list: actions allowed for these reason codes

Stats tools what usually happens, aggregates only

`get_late_rate_by_route`	Historic late rate by area, route, weekday or slot
`get_warehouse_load`	Open pick lines now vs normal for this hour
`get_carrier_performance`	On-time rate and average delay per carrier
`get_action_effectiveness`	How often each action saved the order, per reason code
`get_alert_quality`	Precision@K and dismiss reasons, per model version
`get_kpi_summary`	OTD, flagged and saved orders; same numbers as Power BI

SAME RULES FOR EVERY TOOL

1. **Read-only**
   DB role mcp_reader: SELECT on v_* views only. Writes are done by the worklist app.

2. **No personal data**
   Views hide names, phones, full address and driver IDs.

3. **Scoped and bounded**
   Country from the SSO role; ≤ 50 rows, 3 s, 180 days back.

4. **Fresh and logged**
   as_of on every reply, DATA_STALE after 90 min; every call in tool_call_log.

WHICH TOOLS, WHEN

Verify an alert
`risk_explanation` › `order_context` › `shipment_status` › `warehouse_load`

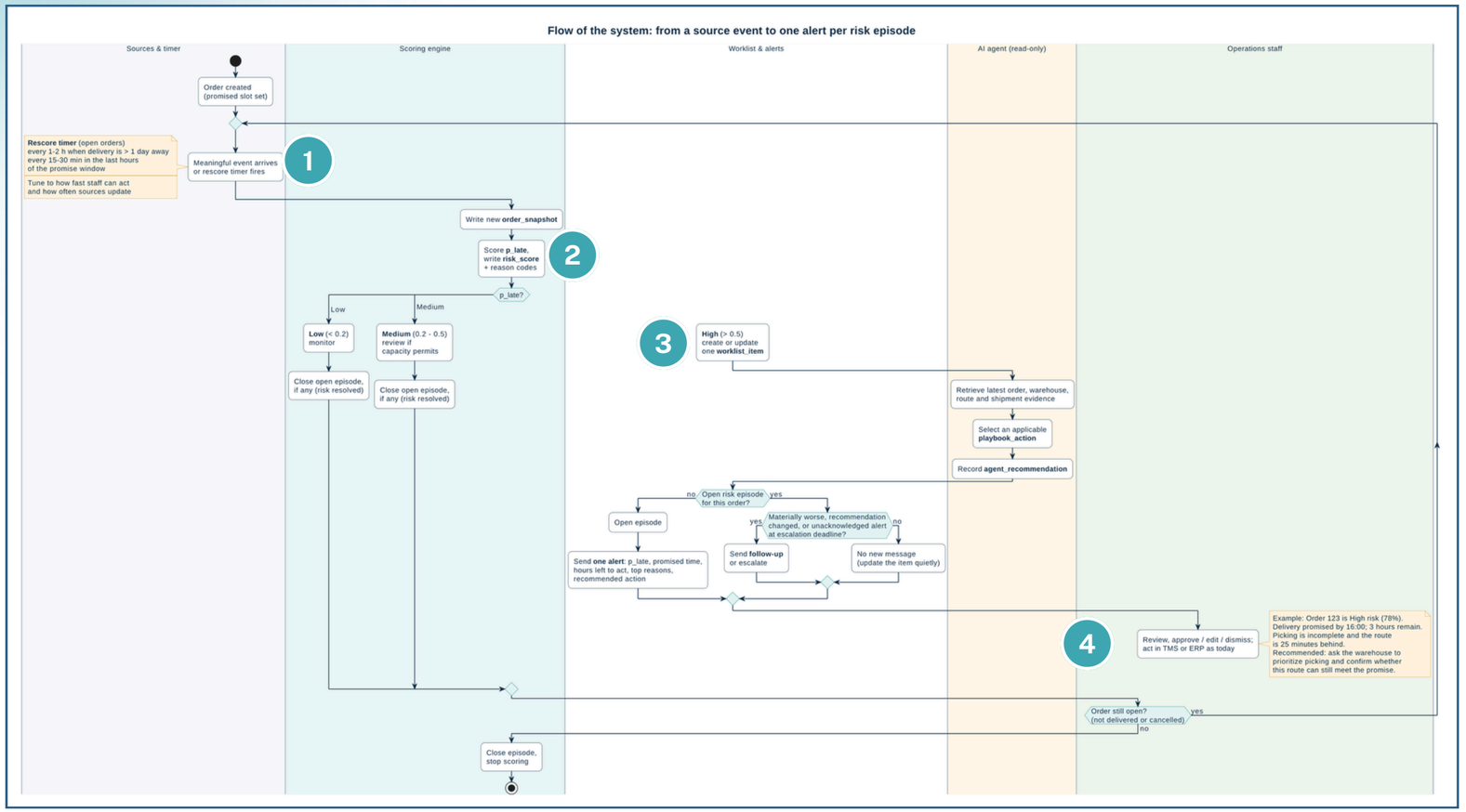
Propose an action
`playbook` › `vehicle_options` / `alt_warehouse_stock` › `similar_past_cases`

Shift briefing
`list_high_risk_orders` › `warehouse_load` › `kpi_summary`

Budget: ≤ 10 calls and 20 s per request. Otherwise the agent returns what it has, marked incomplete, and never guesses.

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Flow & ML engine



1 RESCORE

Every 1–2 h if delivery is > 1 day away, every 15–30 min in the last hours, and on each meaningful event. Silence near the deadline counts too.

2–3 ONE ALERT

High opens an episode: one alert with p_late, promise, hours left, top reasons and the action. Follow up only if it gets worse or nobody acknowledged.

4 DECIDE & CLOSE

Staff approve, edit or dismiss and act in TMS or ERP. The episode closes on delivery, cancellation, or when the risk resolves.

ML ENGINE · TARGET

`risk_score.p_late`

Probability that the order misses its promised window.

Low < 0.2	Medium 0.2–0.5	High > 0.5
monitor	review if capacity	worklist

FEATURES, FROM THE ORDER SNAPSHOT

- **Promise timing** hours left, promised window
- **Order** B2B / B2C, weight, volume, handling
- **Stock & warehouse** short lines, stage, time between events
- **Transport** departure delay, ETA drift, vehicle
- **Destination** region, travel time, holiday, weather
- **Operations** warehouse, carrier, own fleet

EVALUATION

Precision@K (primary): of the K orders planners can review per day, how many were really late? **Precision@K**, **Recall@K** and **PR-AUC** when it is close.

MODEL: COMPARE BOTH

Logistic Regression
baseline, easy to explain

vs

LightGBM
non-linear effects

Same snapshots, tested on a later period. Repeated at every monthly retrain: champion vs challenger.

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MVP and scale: Sweden first

MVP: 12 weeks in Sweden

One delivery region, with staff in control at every step.

- **Weeks 1-3: Validate** — Check assumptions with staff and IT. Collect order and delivery history.
- **Weeks 4-7: Build** — Risk score, prioritised worklist and agent suggestions.
- **Weeks 8-10: Shadow run** — Run next to dispatchers without changing their work.
- **Weeks 11-12: Decide** — Compare with the baseline and decide go or no-go.

In scope

✓ One warehouse and delivery region in Sweden
✓ Risk score per order, warehouse to delivery
✓ Prioritised worklist
✓ Agent suggestions, staff approve
✓ Decisions and outcomes recorded

Not in the MVP

✗ Automatic re-planning
✗ New routing engine
✗ Writing back into ERP or TMS
✗ Norway and Denmark

How we judge success

- Late orders flagged early enough to act
- Dispatchers find the alerts useful
- Less time per delivery exception
- Daily use of the worklist

Scale: one solution, three rollouts



Sweden · Month 0–6

PILOT ONE REGION | ALL OF SWEDEN IF KPI MET

Month 0–3: pilot in one region. Month 3–6: all of Sweden, if the KPI target is met.

Brief priority: delivery and order fulfilment
→ Local focus: site deliveries and crane-truck slots

Norway · Month 6–9 (wave 2)

WAVE 2 NORWAY

Reuse the platform; plug in Norwegian feeds and carriers; calibrate the model.

Brief priority: store operations
→ Local focus: worklist for stores that fulfil orders

Denmark · Month 9–12 (wave 3)

WAVE 3 DENMARK

Same platform with Danish data, carriers, holidays and language.

Brief priority: manual processes
→ Local focus: automate checking and customer updates

Built once vs local

Built once: data model, risk model, worklist, AI agent.

→ Local: connectors, language, risk thresholds, a model retrained on local data, and a dispatcher champion.

→ People: dispatchers co-design the worklist; the tool advises, people decide; alert volume is capped.

Time to market: each new country starts from a working product. Month timings are planning assumptions to confirm with Company A's IT team.

Impact, risks and next steps

Business impact

More deliveries on time · fewer failed or repeat deliveries · less manual checking · fewer “where is my order” calls · stronger trust from B2B customers.

+3 - 5 pts

Target: on-time rate +3–5 pts in the pilot (assumption)
→ Measured: pilot hub vs. control hub over 12 weeks

Validate first

ERP, TMS, Power BI and Google Maps are used as assumed. Promised and actual delivery times are stored per order.

Baseline: current late-delivery rate and its cost in Sweden

Key risks

Late or incomplete data: audit first, start daily if needed. Staff ignore alerts: show reasons, run in shadow.

Too many false alarms: tune alerts to team capacity
→ GDPR: EU hosting, minimal personal data

Our assumptions

ERP/TMS can export open orders every 15–60 min, outcomes are recorded, and dispatchers can act in time.

→ If wrong: start rules-only, add ML when data allows. Owner: product owner + local dispatch lead.

Next 2 weeks

1. Interview dispatchers and IT in the pilot region.
2. Get 6–12 months of order and delivery data.
3. Agree the baseline and go / no-go criteria.
4. Choose the Swedish pilot hub.